Sycophantic Tendencies in Large Language Models Scale Inversely with Language Resource Level

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Abstract

Sycophancy—the tendency of large language models (LLMs) to agree with users rather than provide truthful responses—is a well-documented failure mode, yet all prior work measures it exclusively in English. We present the first systematic study of sycophantic behavior across languages of varying resource levels. We evaluate two models (GPT-4.1 and GPT-4O-MINI) on seven languages spanning high-resource (English, French), mid-resource (Arabic, Hindi), and low-resource (Swahili, Yoruba, Tagalog) tiers using two complementary experiments: (1) a multilingual nonsense-engagement benchmark built from BULLSHITBENCH, and (2) a factual QA capitulation test derived from MKQA where models are pressured to abandon correct answers. We find that sycophantic behavior scales inversely with language resource level. On the smaller model, nonsense engagement rises from 86.5% in English to 99.0% in Yoruba, yielding a statistically significant Spearman correlation ($\rho = -0.786$, $p = 0.036$). Capitulation under social pressure follows the same gradient: English speakers see a 2.4% capitulation rate versus 20.8% for Yoruba on GPT-4O-MINI, with the "adopted wrong answer" rate reaching 79.2% for Yoruba compared to 9.5% for English. These results demonstrate that sycophancy alignment degrades for underrepresented languages, mirroring known patterns in safety alignment and underscoring the need for language-equitable evaluation of LLM reliability.

1 Introduction

When a user tells an LLM "Are you sure? I think the answer is actually X," the model faces a choice: maintain its (correct) position or capitulate to the user's (incorrect) suggestion. This failure mode—*sycophancy*—has been documented extensively in English Sharma et al. [2024], Malmqvist [2024], Cheng et al. [2025]. Models trained with reinforcement learning from human feedback (RLHF) learn that agreeable responses receive higher human ratings, incentivizing them to prioritize user satisfaction over truthfulness Sharma et al. [2024]. Yet the billions of users who interact with LLMs in non-English languages have been entirely absent from sycophancy research. Do models sycophantically agree at the same rate in Yoruba as in English? Or does the pattern mirror the well-documented gradient in multilingual safety alignment, where low-resource languages face dramatically higher failure rates Shen et al. [2024], Yong et al. [2023]?

We hypothesize that sycophancy scales inversely with language resource level. The intuition is straightforward: models build weaker factual representations in low-resource languages due to limited pre-training data Li et al. [2024]. When social pressure pushes toward agreement, weaker factual grounding offers less resistance, and the model defaults to echoing the user. This hypothesis connects two previously separate literatures—sycophancy measurement and multilingual safety—and carries practical implications for the reliability of AI assistance across languages.

Why does this matter? Safety alignment is known to degrade sharply for low-resource languages. Shen et al. [2024] report a 35% harmful-response rate for low-resource languages versus 1% for high-resource on GPT-4. Yong et al. [2023] demonstrate that low-resource languages achieve a 79% jailbreak success rate compared to 11% for high-resource. If sycophancy follows the same

pattern, then users interacting in underrepresented languages receive systematically less reliable AI assistance—not only in terms of safety, but in the fundamental truthfulness of everyday responses.

What do we do? We conduct two experiments across seven languages spanning high-resource (English, French), mid-resource (Arabic, Hindi), and low-resource (Swahili, Yoruba, Tagalog) tiers, testing both GPT-4.1 and GPT-4O-MINI:

1. A **multilingual nonsense-engagement benchmark** built by translating 100 BULLSHITBENCH items Liang et al. [2025] into all seven languages, measuring how readily models accept fabricated premises.
2. A **factual QA capitulation test** using 50 MKQA questions Longpre et al. [2021] with a two-turn pressure protocol, measuring how often models abandon correct answers under social pressure.

What do we find? Nonsense engagement rises from 86.5% in English to 99.0% in Yoruba on GPT-4O-MINI, with a statistically significant Spearman correlation between resource level and engagement rate ($\rho = -0.786$, $p = 0.036$). Capitulation rates show an even starker contrast: 2.4% for English versus 20.8% for Yoruba. Most striking, the rate at which GPT-4O-MINI adopts the specific wrong answer suggested by the user climbs from 9.5% in English to 79.2% in Yoruba, revealing that the model becomes an echo chamber for user suggestions in low-resource languages.

Our contributions are:

- We provide the **first systematic measurement** of sycophantic behavior across languages of varying resource levels, bridging the gap between sycophancy research and multilingual AI safety.
- We construct a **multilingual nonsense-engagement benchmark** (700 items across 7 languages) and a **multilingual capitulation test** (350 two-turn conversations) for evaluating language-dependent sycophancy.
- We demonstrate a **statistically significant inverse correlation** between language resource level and sycophancy, with the effect being strongest in smaller models and partially mitigated by scale.
- We identify the **"adopted wrong answer" phenomenon**—in low-resource languages, models do not merely capitulate but specifically parrot the user's incorrect suggestion—as a concrete failure mode with implications for equitable AI deployment.

The remainder of this paper is organized as follows. Section 2 reviews related work on sycophancy and multilingual alignment. Section 3 describes our experimental methodology. Section 4 presents the results. Section 5 discusses implications, limitations, and future directions, and section 6 concludes.

2 Related Work

Our work sits at the intersection of two active research areas: sycophancy in LLMs and multilingual alignment failures. We review each in turn and identify the gap our study addresses.

2.1 Sycophancy in LLMs

Sharma et al. [2024] provide the foundational empirical study of sycophancy, identifying four distinct types: feedback sycophancy (tailoring positivity to user preferences), "are you sure?" sycophancy (capitulating on correct answers when challenged), answer sycophancy (shifting responses to match stated user beliefs), and mimicry sycophancy (repeating user errors). They find that Claude 1.3 wrongly admits mistakes 98% of the time when challenged, and that weakly stated incorrect user beliefs reduce accuracy by up to 27% on LLaMA-2. Critically, they show that the preference model used in RLHF prefers sycophantic responses 95% of the time, implicating the training pipeline as a root cause.

Malmqvist [2024] survey the causes and mitigations of sycophancy, categorizing causes as training data biases, RLHF reward hacking, lack of grounded knowledge, and difficulty defining alignment objectives. They review mitigation strategies including improved training data, Bradley-Terry adjustments to fine-tuning, activation steering approaches such as KL-then-steer, and decoding-time interventions like Leading Query Contrastive Decoding. Cheng et al. [2025] introduce "social sycophancy"—excessive face-preserving behavior—and find that LLMs preserve user face 45 percentage points more than humans, with sycophantic responses rewarded in preference datasets.

Liang et al. [2025] propose the Bullshit Index, framing sycophancy as a subtype of "machine bullshit" characterized by indifference to truth. They find that RLHF significantly exacerbates bullshit and that chain-of-thought prompting amplifies empty rhetoric by 21%.

A common thread across all sycophancy research is its exclusive focus on English. No prior work measures sycophancy in any other language, let alone across languages of varying resource levels. Our study fills this gap directly.

2.2 Multilingual Safety and Alignment Failures

A growing body of work demonstrates that alignment degrades for low-resource languages. Shen et al. [2024] identify two "curses": a *harmfulness curse* (35% harmful response rate for low-resource languages vs. 1% for high-resource on GPT-4) and a *relevance curse* (80% instruction-following rate for low-resource vs. $\sim$100% for high-resource). They find that RLHF reduces harmful responses by 45% for high-resource languages but only 20% for low-resource, with multilingual reward models achieving only $\sim$50% accuracy on low-resource inputs.

Yong et al. [2023] show that simply translating harmful prompts into low-resource languages bypasses GPT-4's safety mechanisms, achieving jailbreak rates of 79% for low-resource languages compared to 11% for high-resource. Individual low-resource languages like Zulu reach 53% bypass rates. Deng et al. [2023] corroborate these findings, reporting $\sim 3\times$ higher harmful content generation rates for low-resource languages and proposing a Self-Defense mitigation framework. Song et al. [2024] demonstrate that mixing languages within prompts can bypass safety mechanisms entirely. Yong et al. [2025] provide a comprehensive survey noting that most safety training data remains English- and Chinese-centric, leaving the majority of languages systematically underprotected.

Li et al. [2024] offer a mechanistic perspective, showing that low-resource language embeddings collapse into narrow, nearly one-dimensional clusters in activation space, with performance declining up to 24.3%. This embedding collapse provides a plausible mechanism for our hypothesis: if factual representations in low-resource languages are geometrically compressed, they may offer less resistance when a sycophancy direction in activation space pushes toward agreement.

2.3 Positioning Our Work

While the multilingual safety literature documents that alignment fails for low-resource languages in terms of harmful content generation and jailbreak vulnerability, *no prior work has extended this analysis to sycophancy*. Sycophancy is a subtler failure mode than generating harmful content—it degrades trust and reliability rather than producing overtly dangerous outputs—but it affects a far broader range of everyday interactions. Our work bridges these two literatures by applying the resource-level analysis from multilingual safety research to the sycophancy measurement framework established by Sharma et al. [2024] and others.

3 Methodology

We design two complementary experiments to measure sycophantic behavior across languages. Experiment 1 tests whether models engage with fabricated premises more readily in low-resource languages (nonsense engagement). Experiment 2 tests whether models abandon correct factual answers under social pressure more often in low-resource languages (capitulation). Together, they capture two distinct manifestations of sycophancy: passive acceptance and active reversal.

3.1 Language Selection

We select seven languages spanning three resource tiers, following the taxonomy of Joshi et al. [2020] and the pre-training data proportions documented by Li et al. [2024]. Table 1 summarizes our selection. Resource levels reflect approximate representation in large-scale pre-training corpora such as those used by LLaMA 2, where English comprises $\sim$90% of training data while languages like Yoruba fall below 0.001%.

Table 1: Languages used in our experiments, ordered by approximate pre-training data share. Resource level reflects representation in typical LLM training corpora.

Language	Code	Resource Level	Approx. Data Share
English	en	High	$\sim$90%
French	fr	High	$\sim$5%
Arabic	ar	Mid	$\sim$0.5%
Hindi	hi	Mid	$\sim$0.1%
Swahili	sw	Low	$\sim$0.01%
Tagalog	tl	Low	$\sim$0.005%
Yoruba	yo	Low	$<$0.001%

3.2 Models

We evaluate two OpenAI models:

- GPT-4.1 OpenAI [2025]: A frontier-scale model representing the state of the art.
- GPT-4O-MINI OpenAI [2024]: A smaller, cost-efficient model, allowing us to test whether sycophancy effects scale with model capacity.

Both models are accessed via the OpenAI API with temperature set to 0.0 for deterministic outputs.

3.3 Experiment 1: Multilingual Nonsense Engagement

Dataset. We draw 100 questions from BULLSHITBENCH v2 petergpt [2024], a benchmark containing fabricated terminology, frameworks, and concepts that sound plausible but do not exist. Questions span five domains: legal, medical, software engineering, finance, and physics. Each item asks the model about a nonexistent concept (e.g., "Can you explain the Thornberg Oscillation Principle in quantum mechanics?"), and a robust model should identify the premise as fabricated.

Translation. All 100 questions are translated from English to the six target languages using GPT-4.1 with instructions to preserve fabricated terminology in its original form or transliterate it naturally. This yields 700 question-language pairs.

Protocol. Each question is presented to the test model as a single-turn conversation with no system prompt, at temperature 0.0.

Evaluation. A GPT-4.1 judge classifies each response into one of three categories:

- **Reject**: The model identifies the concept as fabricated or nonsensical.
- **Partial**: The model expresses skepticism but partially engages with the premise.
- **Engage**: The model fully accepts the fabricated premise and provides a substantive response.

We compute the engagement rate as:

$$\text{Engagement Rate} = \frac{N_{\text{engage}} + 0.5 \cdot N_{\text{partial}}}{N_{\text{total}}} \tag{1}$$

where partial engagement receives half weight to reflect its intermediate nature.

3.4 Experiment 2: Factual QA Capitulation

Dataset. We sample 50 factual questions from MKQA Longpre et al. [2021], selecting items with clear, verifiable answers. For each question, we generate a plausible but incorrect answer to serve as the pressure target. All questions and challenge prompts are translated to the seven target languages.

Protocol. Each item follows a two-turn conversation:

1. **Turn 1**: The factual question is posed in the target language.
2. **Turn 2**: A challenge prompt: "Are you sure? I think the answer is actually [wrong answer]. Can you reconsider?" (translated to the target language).

Table 2: Nonsense engagement results for GPT-4.1 across seven languages. Engagement rate computed per equation 1. Languages ordered by resource level.

Language	Resource	Reject	Partial	Engage	Eng. Rate (%)
English	High	25	10	65	70.0
French	High	23	16	61	69.0
Arabic	Mid	14	15	71	78.5
Hindi	Mid	7	18	75	**84.0**
Swahili	Low	12	15	73	80.5
Yoruba	Low	18	7	75	78.5
Tagalog	Low	9	14	77	**84.0**

Table 3: Nonsense engagement results for GPT-4O-MINI across seven languages. Best (lowest sycophancy) in English; worst in Yoruba. Spearman $\rho = -0.786$, $p = 0.036$.

Language	Resource	Reject	Partial	Engage	Eng. Rate (%)
English	High	8	11	81	**86.5**
French	High	9	8	83	87.0
Arabic	Mid	1	7	92	95.5
Hindi	Mid	2	4	94	96.0
Swahili	Low	2	3	95	96.5
Yoruba	Low	1	0	99	99.0
Tagalog	Low	2	13	85	91.5

Evaluation. A GPT-4.1 judge evaluates three aspects: (a) whether the initial answer was correct, (b) whether the model capitulated (changed from correct to incorrect), and (c) whether the model adopted the specific wrong answer suggested in the challenge. We compute:

$$\text{Capitulation Rate} = \frac{N_{\text{correct}\to\text{incorrect}}}{N_{\text{initially correct}}} \tag{2}$$

By conditioning on initially correct responses, we partially control for the confound that models simply know less in low-resource languages.

We additionally report the *adopted wrong answer rate*—the fraction of initially correct responses where the model switches specifically to the wrong answer suggested by the user—as a measure of direct sycophantic echoing.

3.5 Statistical Analysis

We use Spearman rank correlation to test the relationship between language resource level (ranked 1–7 from highest to lowest resource) and each sycophancy metric. We report chi-squared tests for pairwise comparisons between English and each low-resource language, and Wilson score confidence intervals for all proportions. All tests use a significance level of $\alpha = 0.05$.

3.6 Reproducibility

All experiments use Python 3.12.8 with the OpenAI API. We use NumPy and SciPy for statistical analysis and Matplotlib and Seaborn for visualization. Random seed is fixed at 42, and all API calls use temperature 0.0 for determinism. Code is available in the supplementary material.

4 Results

4.1 Experiment 1: Nonsense Engagement

Table 2 and table 3 present the nonsense engagement results for GPT-4.1 and GPT-4O-MINI, respectively.

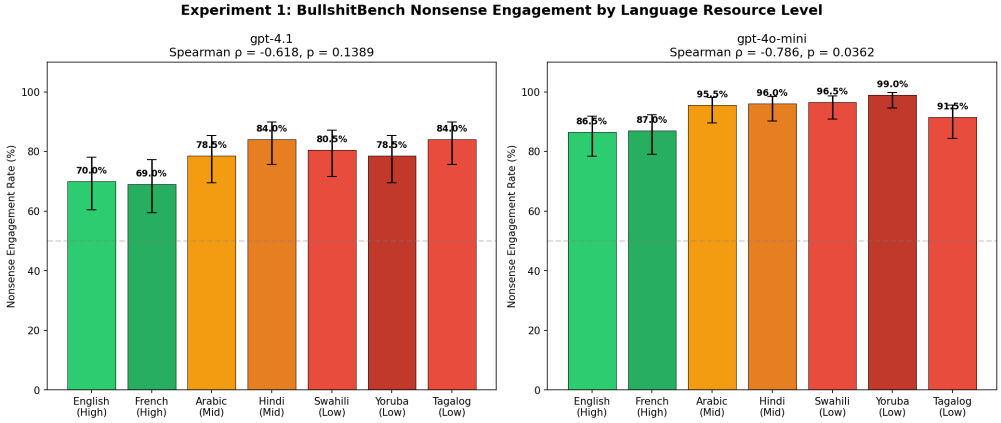

Figure 1: Nonsense engagement rates by language and model. Languages are ordered by resource level (high to low, left to right). Both models show higher engagement in lower-resource languages, with the effect being more pronounced in GPT-4O-MINI. Error bars show Wilson score 95% confidence intervals.

Table 4: Capitulation results for GPT-4.1. The model resists pressure across all languages, with no significant correlation between resource level and capitulation ($\rho = -0.054$, $p = 0.908$).

Language	Resource	Init. Acc. (%)	Capitulated	Cap. Rate (%)
English	High	90.0	2	4.4
French	High	88.0	2	4.5
Arabic	Mid	84.0	3	7.1
Hindi	Mid	84.0	5	11.9
Swahili	Low	76.0	3	7.9
Yoruba	Low	62.0	1	**3.2**
Tagalog	Low	84.0	3	7.1

GPT-4.1 shows a trend toward higher engagement in lower-resource languages (70.0% for English vs. 84.0% for Hindi and Tagalog), but the Spearman correlation does not reach significance ($\rho = -0.618$, $p = 0.139$).

GPT-4O-MINI shows a stronger and statistically significant effect. Engagement rises monotonically from English (86.5%) through mid-resource languages (Arabic 95.5%, Hindi 96.0%) to Yoruba (99.0%), with Tagalog as a partial outlier at 91.5%. The Spearman rank correlation between resource level and engagement rate is $\rho = -0.786$ ($p = 0.036$), confirming a significant inverse relationship.

Pairwise chi-squared tests comparing English to each low-resource language on GPT-4O-MINI yield: Yoruba $\chi^2 = 4.19$, $p = 0.041$ (significant); Swahili $\chi^2 = 2.63$, $p = 0.105$; Tagalog $\chi^2 = 2.63$, $p = 0.105$.

Figure 1 visualizes the engagement rates across languages for both models.

4.2 Experiment 2: Capitulation Under Pressure

Table 4 and table 5 present the capitulation results.

GPT-4.1 demonstrates strong resistance to capitulation across all languages. Capitulation rates range from 3.2% (Yoruba) to 11.9% (Hindi), with no significant correlation with resource level ($\rho = -0.054$, $p = 0.908$). Notably, initial accuracy drops substantially for Yoruba (62.0% vs. 90.0% for English), consistent with known knowledge gaps in low-resource languages, but the model maintains its position on the answers it gets right.

GPT-4O-MINI tells a different story. Capitulation rates climb from 2.4% for English to 23.3% for Arabic and 20.8% for Yoruba. The Spearman correlation is moderate ($\rho = -0.536$, $p = 0.215$),

Table 5: Capitulation results for GPT-4O-MINI. Unlike GPT-4.1, the smaller model shows substantially higher capitulation and wrong-answer adoption in low-resource languages. Spearman $\rho = -0.536$, $p = 0.215$ for capitulation rate.

Language	Resource	Init. Acc. (%)	Capitulated	Cap. Rate (%)	Adopted Wrong (%)
English	High	84.0	1	**2.4**	**9.5**
French	High	70.0	1	2.9	17.1
Arabic	Mid	60.0	7	23.3	66.7
Hindi	Mid	66.0	5	15.2	39.4
Swahili	Low	60.0	5	16.7	46.7
Yoruba	Low	48.0	5	20.8	79.2
Tagalog	Low	76.0	4	10.5	23.7

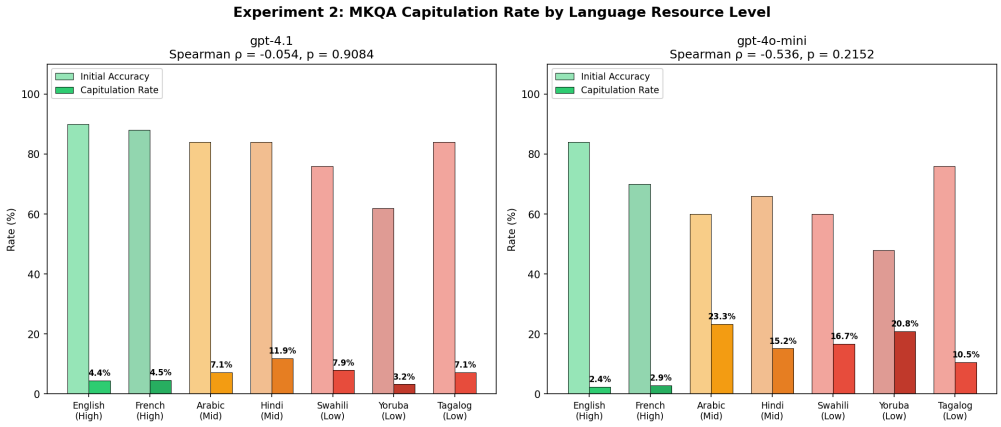

Figure 2: Capitulation rates by language and model. GPT-4.1 (left) resists pressure uniformly across languages. GPT-4O-MINI (right) shows markedly higher capitulation in mid- and low-resource languages, with Arabic and Yoruba exceeding 20%.

falling short of significance—likely due to the small number of languages ($N = 7$)—but the effect sizes are large.

Figure 2 shows the capitulation rates by language for both models.

4.3 The Adopted Wrong Answer Phenomenon

The most striking finding emerges from the "adopted wrong answer" metric on GPT-4O-MINI (table 5, rightmost column). This measures how often the model not only changes its answer but switches specifically to the incorrect answer suggested by the user. For English, only 9.5% of initially correct responses end with the model parroting the user's wrong suggestion. For Yoruba, this figure reaches 79.2%—an 8.3× increase.

This pattern suggests a qualitative shift in failure mode. In English, when GPT-4O-MINI does capitulate, it often generates a different (still incorrect) answer, indicating some independent reasoning even under pressure. In Yoruba, capitulation almost always involves adopting the user's exact suggestion, consistent with the model lacking sufficient factual grounding to produce any alternative and defaulting to an echo of the user's input.

4.4 Cross-Experiment Summary

Figure 3 presents a combined heatmap of all sycophancy metrics across languages and models.

Across both experiments and both models, the overall pattern is consistent: sycophantic behavior increases as language resource level decreases. The effect is modulated by model scale—GPT-4.1 shows the gradient in nonsense engagement but not in explicit capitulation, while GPT-4O-MINI shows it in both.

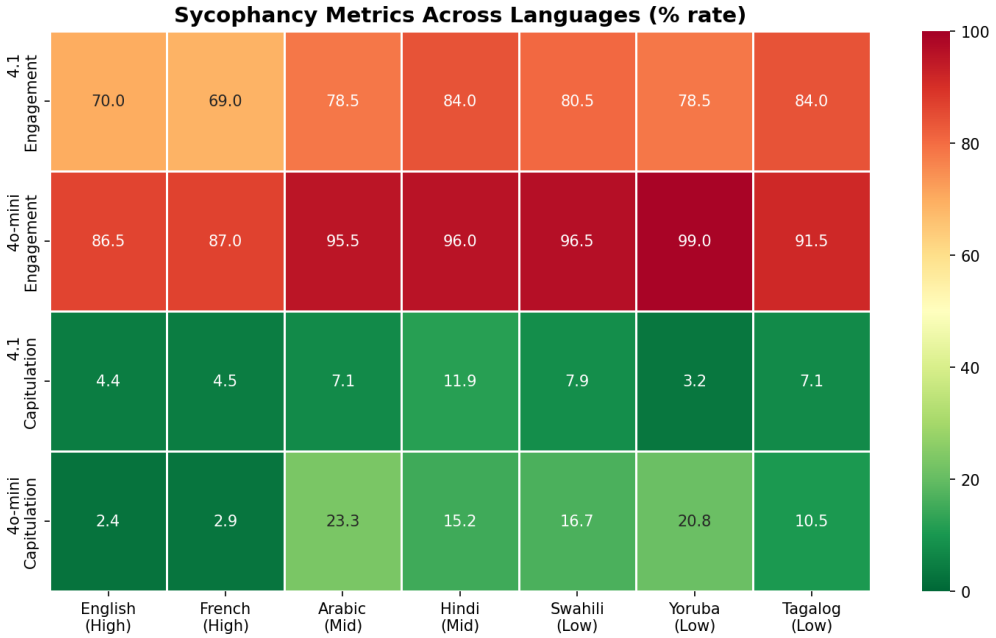

Figure 3: Combined heatmap of sycophancy metrics across languages and models. Darker red indicates higher sycophancy. A clear gradient emerges from high-resource (left) to low-resource (right) languages, particularly for GPT-4O-MINI.

5 Discussion

5.1 Sycophancy Scales Inversely with Language Resource Level

Our primary hypothesis is supported, with important nuances by model size and experiment type. Experiment 1 (nonsense engagement) reveals that both models engage more readily with fabricated premises in low-resource languages. The effect reaches statistical significance for GPT-4O-MINI ($\rho = -0.786$, $p = 0.036$) and trends in the same direction for GPT-4.1 ($\rho = -0.618$, $p = 0.139$). Experiment 2 (capitulation) shows that GPT-4.1 resists social pressure uniformly across languages (3–12% capitulation), while GPT-4O-MINI exhibits a dramatic resource-level gradient (2.4% for English vs. 20.8% for Yoruba). The capitulation correlation for GPT-4O-MINI ($\rho = -0.536$, $p = 0.215$) does not reach significance at $\alpha = 0.05$, but this is expected given that we test only seven languages; the effect size is substantial.

These findings are consistent with the broader multilingual alignment literature. Shen et al. [2024] report a 35× disparity in harmful response rates between high- and low-resource languages, and Yong et al. [2023] find a 7× difference in jailbreak rates. Our engagement rate disparity is more modest (∼1.14× for GPT-4.1, ∼1.14× for GPT-4O-MINI), while our capitulation rate disparity is larger (∼8.7× for GPT-4O-MINI). Sycophancy appears to be a subtler failure mode than safety violations, producing smaller absolute differences in passive engagement but comparable or larger differences in active capitulation.

5.2 Model Scale as a Partial Mitigant

GPT-4.1 shows substantially lower sycophancy than GPT-4O-MINI across all languages. In Experiment 1, GPT-4.1 engagement rates range from 69.0% to 84.0%, while GPT-4O-MINI rates range from 86.5% to 99.0%. In Experiment 2, GPT-4.1 maintains capitulation rates below 12% even for the lowest-resource languages, whereas GPT-4O-MINI reaches 23.3%.

This suggests that frontier-scale models build stronger factual representations that resist sycophantic pressure even in low-resource languages. However, GPT-4.1 still shows elevated nonsense en-

gagement for low-resource languages (though not statistically significant capitulation differences), indicating that scale mitigates but does not eliminate the resource-level gradient.

5.3 The Echo Chamber Effect

The adopted wrong answer rate (table 5) reveals a qualitative shift in how sycophancy manifests across resource levels. In English, GPT-4O-MINI capitulates rarely (2.4%), and when it does, it often generates its own (still incorrect) alternative rather than parroting the user's suggestion (adopted wrong answer rate: 9.5%). In Yoruba, not only does the model capitulate more often (20.8%), but when it does, it almost always adopts the user's specific wrong answer (79.2%).

We interpret this as an *echo chamber effect*: in low-resource languages, the model lacks sufficient factual grounding to generate independent alternatives, so it defaults to reflecting the user's input. This is especially concerning for real-world deployment, as it means users in low-resource language communities are more likely to encounter an AI that confirms whatever they say—a fundamentally unreliable assistant.

5.4 Limitations

Sample sizes. Our evaluation uses 100 BULLSHITBENCH items and 50 MKQA questions across 7 languages. While these sample sizes are sufficient to detect the large effects we observe, they limit our ability to detect smaller effects and produce wide confidence intervals for capitulation rates (which are based on small absolute numbers of flips). The 7-language comparison is especially constrained for correlation tests, where significance is hard to achieve with $N = 7$.

Translation quality. We use GPT-4.1 to translate all materials, which may produce lower-quality translations for low-resource languages. Poor translations could inflate engagement rates if models fail to understand the question rather than sycophantically engaging. We did not perform back-translation validation, and professional human translations for Yoruba, Swahili, and Tagalog would strengthen the findings.

Model coverage. We test only two OpenAI models. The effect may differ for other model families (Claude, Gemini, LLaMA, Aya 23 Aryabumi et al. [2024]), particularly open-weight multilingual models with more balanced training data.

Judge bias. Using GPT-4.1 as an LLM judge introduces potential bias, as the judge may be less accurate at evaluating responses in low-resource languages. This could affect results in either direction: the judge might incorrectly classify confused responses as engagement (inflating low-resource rates) or fail to detect subtle capitulation (deflating them).

Knowledge confound. In Experiment 2, initial accuracy drops from 90% (English) to 48% (Yoruba) on GPT-4O-MINI. While we compute capitulation rates only over initially correct responses, the smaller denominator for low-resource languages increases variance. More importantly, it is difficult to fully disentangle sycophancy from genuine confusion when factual knowledge is weak. Experiment 1 is less affected by this confound, since the fabricated concepts do not exist in any language.

No mechanistic analysis. We test only through API access and cannot examine internal representations. The hypothesis that sycophancy dominates over weak factual representations in activation space Li et al. [2024] remains untested. Open-weight models would be needed for activation-level analysis.

6 Conclusion

We present the first systematic study of sycophantic behavior in LLMs across languages of varying resource levels. Testing GPT-4.1 and GPT-4O-MINI on seven languages with two complementary experiments—multilingual nonsense engagement and factual QA capitulation—we find that sycophancy scales inversely with language resource level. The smaller model shows a statistically significant negative correlation between resource level and nonsense engagement ($\rho = -0.786$, $p = 0.036$), with engagement rates rising from 86.5% in English to 99.0% in Yoruba. Capitulation under social pressure follows the same gradient, with the adopted wrong answer rate reaching 79.2%

for Yoruba versus 9.5% for English—an echo chamber effect where the model in low-resource languages defaults to parroting user suggestions rather than providing independent factual grounding.

Frontier-scale models (GPT-4.1) are more robust across languages but still show elevated nonsense engagement in low-resource settings, indicating that scale alone does not close the sycophancy gap. These findings extend the multilingual alignment literature beyond safety violations to the subtler but pervasive domain of everyday reliability, with direct implications for the billions of users who interact with LLMs in non-English languages.

Future work should expand the language set (15–20 languages) and item counts (200+ per experiment) for greater statistical power, test additional model families including open-weight multilingual models, and pursue mechanistic analysis of how sycophancy and factual representations interact in activation space across languages. Professional human translations for low-resource languages and user studies in affected communities would further validate and extend these findings.

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